

CSCE 679 DATA VISUALIZATION · MODULE 12 · LECTURE NOTES

Scientific Visualisation

Fields, volumes, and the return to rendering.

MODULE OUTCOMES

- Distinguish scientific from information visualisation.
- Explain scalar field techniques and the transfer function.
- Explain volume rendering and its connection to CSCE 647.
- Explain vector and tensor field visualisation.
- Choose a technique for a field and a question.

1 The difference

IN SCIENTIFIC VISUALISATION THE POSITIONS ARE GIVEN BY THE DATA, SO THE DESIGN QUESTION IS DIFFERENT

In information visualisation you choose where the marks go — which is the entire subject of Modules 01 through 07 — **and the position channel, rank one, is yours to assign** to whichever variable most needs it.

In scientific visualisation the positions are given by the physics — a temperature field on a grid, a flow through a domain, a medical scan — **so position is already used, and the design question becomes what to do with the remaining channels.**

Which means colour, opacity, and derived geometry carry everything — and **Module 04's colour rules become the central design decision** rather than one consideration among several, which is why the rainbow scale has done so much damage specifically in the sciences (Module 04 §2).

And the central problem is occlusion: a three-dimensional field has interior structure, and anything in front of it hides what is behind — which is §3's subject and **is exactly what volume rendering exists to solve.**

2 Scalar fields

2D FIELDS

colour map	the whole field at once, with Module 04's rules
contour lines	specific level sets, precisely
both together	the standard, and usually the right answer

3D FIELDS

slices	a 2D field at a chosen plane. Simple, exact, and it shows only that plane.
isosurfaces	the surface where the field equals a chosen value. Shows one level crisply and hides everything else.
direct volume rendering	accumulate colour and opacity along a ray through the whole volume (section 3)

AND THE ISOSURFACE'S THRESHOLD IS THE WHOLE PICTURE.
Change it and the apparent structure changes, which
makes it the same kind of undisclosed analytical
choice as a bin size (Module 11 section 2).

An isosurface's threshold is the whole picture — which makes it the same kind of undisclosed analytical choice as a bin size (Module 11 §2) or a colour binning (Module 08 §3). **The threshold-is-a-choice point connects this module to the rest of the course:** the techniques are different and the honesty requirement is identical, which is to state the parameter and show the figure at two or three values of it before trusting a feature.

CONTOURS AND COLOUR TOGETHER, BECAUSE THEY ANSWER DIFFERENT QUESTIONS

A colour map shows the whole field at once and reads the individual values imprecisely — rank six (Module 02 §1) — **which suits seeing the overall structure** and does not suit reading a value.

Contour lines show specific levels exactly, because a labelled contour is a position-encoded value — you read it by finding the line — **and they show nothing at all between the levels.**

So the combination gives the overall shape and the precise levels together, which is **why topographic maps have used both for two centuries** and is a good example of a combination arrived at empirically long before the perceptual argument for it existed.

And the contour spacing is itself a choice — **evenly spaced contours convey the gradient through their density:** closely spaced lines mean a steep gradient, which the reader perceives directly. **That is an elegant free channel, and it is lost entirely if the spacing is irregular,** so irregular contour intervals should be a deliberate and stated choice.

3 Volume rendering

$$\mathbf{C} = \int \mathbf{c}(\mathbf{s}) \cdot \alpha(\mathbf{s}) \cdot \exp\left(-\int_0^s \alpha(\mathbf{t}) \, dt\right) \, ds$$

- **Accumulate the emitted colour along the ray, attenuated by the total opacity in front of it** — which is the emission-absorption model, and is the standard formulation.
- **Discretised, it becomes front-to-back compositing:** $C \approx \sum_i c_i \alpha_i \prod_{j<i} (1 - \alpha_j)$ — which is exactly the same recurrence as alpha blending, and is why the implementation is a familiar one.
- **And this is CSCE 647's participating-media integral** — the same equation, with a measured scalar field in place of a physical density and a chosen transfer function in place of a material's optical properties.
- **Making the CSCE 647 connection explicit is the point of this module within this track:** the ray marching, the step size and its aliasing, the early termination when the accumulated opacity saturates, and the acceleration structures are all the same material, applied to a different kind of input (§4's closing callout).

AND THE TRANSFER FUNCTION IS THE DESIGN, WHICH MAKES IT THE HARD PART

The transfer function maps the scalar value at each sample to a colour and an opacity — and it determines what is visible, what is transparent, and therefore what the resulting image shows at all.

Which means the picture is a function of a choice nobody looking at it can see: two different transfer functions on the same volume produce images with visibly different apparent structures, both of them faithful renderings of identical data.

And designing one is genuinely hard — it is a mapping from a one-dimensional value to four output channels, with the opacity's effect being non-local because of the accumulation: raising the opacity of one value range hides everything behind it, which is not obvious from the function itself.

So the honest requirement is to publish the transfer function alongside the image — which is this module's version of Module 11 §4's four statements, and is equally rarely done even in published scientific work where the image is the evidence.

4 Vector and tensor fields

- **Arrow glyphs** — direction and magnitude at each sample point. **Simple, immediately understood, and it clutters at once, and it has a sampling artefact: the glyph positions are a grid you chose**, and the reader sees the grid as well as the field.
- **Streamlines** — curves everywhere tangent to the field — which **show the structure far better than glyphs and lose the magnitude entirely** unless it is encoded on colour or on line width.
- **Line integral convolution** — a noise texture smeared along the field direction — which **shows the direction everywhere and densely, is genuinely beautiful, and is hard to read quantitatively** since there are no discrete marks to measure.

- **And seed placement is the design decision for streamlines — uniform seeds give uneven coverage, because the field itself concentrates them:** streamlines converge where the flow converges, leaving bundles and gaps.
- **With tensor fields harder still**, usually reduced either to glyphs (ellipsoids, superquadrics) or to derived scalar fields that are then visualised by §2's methods. **Seed placement is the streamline design — uniform seeds produce dense bundles and empty regions, which is a picture of the seeding rather than of the field**, and the adaptive placement algorithms exist for exactly that reason.

AND THE CONNECTION BACK TO THE GRAPHICS TRACK

Volume rendering is CSCE 647's ray marching, isosurface extraction is a geometry problem that CSCE 620's computational geometry addresses (marching cubes and its successors), **and the whole pipeline is a rendering pipeline** with a measured field as its scene.

Which means this module's techniques are implemented with the tools of that track — and the performance work is literally the same work: acceleration structures, empty-space skipping, early ray termination, and GPU parallelism, all from CSCE 647 and CSCE 645.

And the difference is entirely in the objective: CSCE 647 wants the image to be physically correct, and this module wants it to be *readable* — which is a different and sometimes directly conflicting goal.

So a scientifically useful volume rendering may look physically wrong on purpose — exaggerated opacity, non-physical colour, clipped regions, and discontinuous transfer functions — and that is correct, because the objective is communication rather than simulation. Which is a satisfying place for this track's rendering material to arrive.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Engel et al. — Real-Time Volume Graphics	§3 in full — the integral, the discretisation, and the GPU implementation. Library copy.
Levoy — Display of Surfaces from Volume Data (free)	§3 in the original — direct volume rendering as it was introduced, and the transfer function argument is already there.
Lorensen & Cline — Marching Cubes (free)	§2's isosurface extraction — and the connection to CSCE 620's geometry.
Cabral & Leedom — Line Integral Convolution (free)	§4's third technique in the original , with the construction explained.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State what is different** about the design problem when position is given.
- 2 **Render a 2D scalar field** as colour, as contours, and as both.
- 3 **Vary the contour spacing** and report what the density conveys.
- 4 **Extract an isosurface at three thresholds** and compare the structures.
- 5 **Implement direct volume rendering by ray marching**, reusing your CSCE 647 code.
- 6 **Show that the discretised integral is alpha blending**.
- 7 **Design two transfer functions** for the same volume that suggest different structures.
- 8 **Publish the transfer function** alongside one of your images.
- 9 **Render a vector field as glyphs, streamlines, and LIC**, and compare what each shows.
- 10 **Seed streamlines uniformly, then adaptively**, and compare the coverage.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What is different about the design problem here?
- 2 Why does colour become the central decision?
- 3 What is the central problem of 3D fields?
- 4 Name three 2D and three 3D scalar techniques.
- 5 Why is an isosurface's threshold like a bin size?
- 6 Why are contours and colour used together, and what does contour density convey?
- 7 Give the volume rendering integral and its discretisation.
- 8 Which CSCE 647 equation is it?
- 9 Why is the transfer function the hard part, and what follows?
- 10 Name three vector field techniques and what each loses.